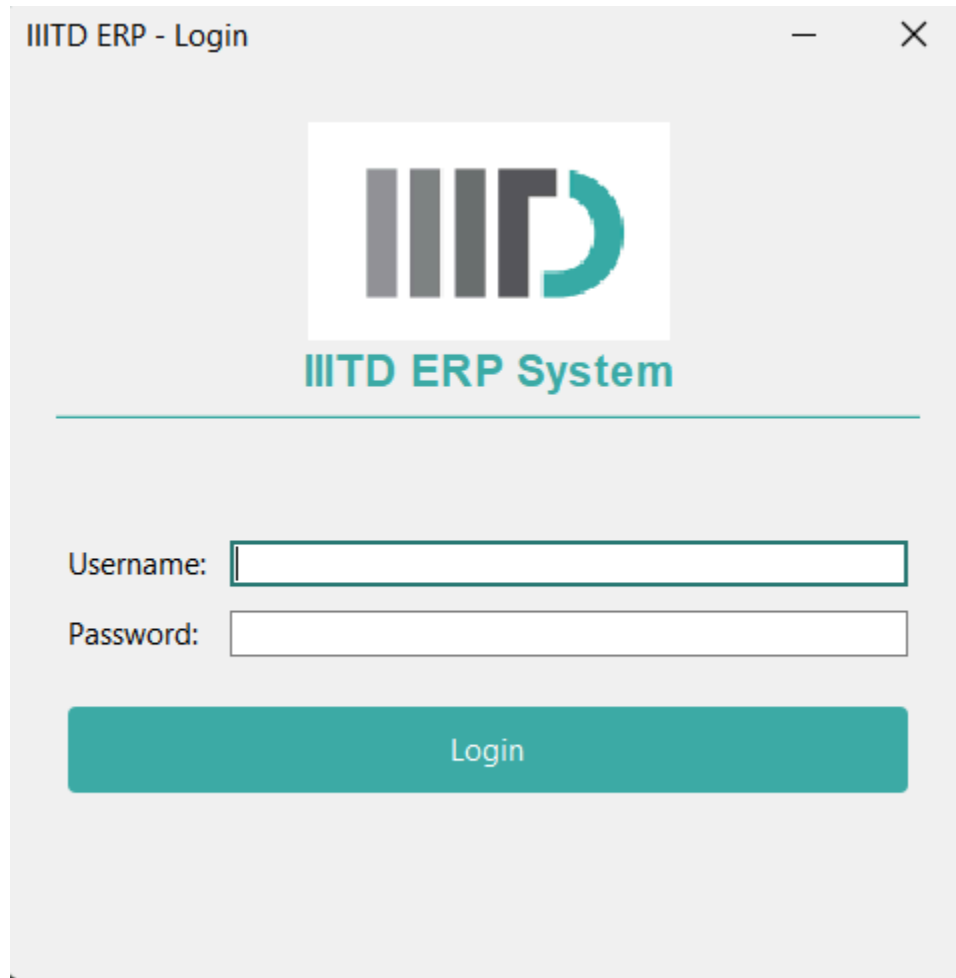


AP PROJECT 2025

REPORT

Screenshots

Login Screen



The screenshot shows a web browser window titled "IIITD ERP - Login". The window contains the IIITD logo, which consists of four vertical bars of increasing height followed by a teal semi-circle. Below the logo, the text "IIITD ERP System" is displayed. A horizontal teal line separates the header from the login fields. There are two input fields: "Username:" and "Password:". Below these fields is a teal "Login" button.

IIITD ERP - Login

IIITD ERP System

Username:

Password:

Login

Admin Dashboard

University ERP - Admin Dashboard

Logout

INDRAPRASTHA INSTITUTE of
INFORMATION TECHNOLOGY DELHI

Welcome, admin (Admin)

User Management

Add Instructor

Add Student

Manage Users

Course & Section Management

Create Course

Create Section

Assign Instructor

Manage Courses

Manage Sections

Maintenance Mode: ☐ OFF

Status messages will appear here...

Instructor Dashboard

University ERP - Instructor Dashboard

Logout

INDRAPRASTHA INSTITUTE of
INFORMATION TECHNOLOGY DELHI

Welcome, Jalote (Instructor)

My Sections

Section ID	Course Code	Course Title	Semester	Year	Day/Time	Room
6	CSE101	Introduction to Progr... 1		2024	MON 09:00-10:30;WE...	C102

Open Gradebook

Refresh

Student Dashboard

University ERP - Student Dashboard

Welcome, 2024419 - Prajeet (Student) [Logout](#)

[Catalog](#) [My Registrations](#) [Timetable](#) [Grades](#) [Export Transcript](#)

Course Catalog

Course Co...	Title	Credits	Section ID	Instructor...	Capacity	Enrolled	Seats Left	Day/Time	Semester	Year
CSE101	Introductio...	4	6	Jalote	2	2	0	MON 09:00-...	1	2024

[Refresh](#) [Register in Selected Section](#)

Table Lists for the Databases

AUTH_DB — Authentication Database

1. users_auth

Columns

- user_id — INT, PK, AUTO_INCREMENT
- username — VARCHAR(64), UNIQUE, NOT NULL
- role — ENUM('student', 'instructor', 'admin'), NOT NULL
- password_hash — VARCHAR(255), NOT NULL
- status — VARCHAR(20), DEFAULT 'active'
- last_login — DATETIME

Indexes

- idx_username
- idx_role

2. auth_failed_logins *(optional — for security monitoring)*

Columns

- username — VARCHAR(64), PK
- failed_count — INT, DEFAULT 0
- last_failed — DATETIME

Index

- idx_last_failed

ERP_DB — ERP Data

1. students

Columns

- user_id — INT, PK (ref: auth_db.users_auth.user_id)
- roll_no — VARCHAR(30)
- program — VARCHAR(100)
- year — INT

Indexes

- idx_roll_no
- idx_program

2. instructors

Columns

- user_id — INT, PK (ref: users_auth.user_id)
- department — VARCHAR(100)

Index

- idx_department

3. courses

Columns

- course_id — INT, PK, AUTO_INCREMENT
- code — VARCHAR(20), UNIQUE, NOT NULL
- title — VARCHAR(255), NOT NULL
- credits — INT, NOT NULL

Index

- idx_code

4. sections

Columns

- section_id — INT, PK, AUTO_INCREMENT
- course_id — INT, NOT NULL
- instructor_id — INT, NULL
- day_time — VARCHAR(100)
- room — VARCHAR(50)
- capacity — INT, DEFAULT 0, CHECK >= 0
- semester — VARCHAR(20)
- year — INT

Indexes

- idx_course_id
- idx_instructor_id
- idx_semester_year

5. enrollments

Columns

- enrollment_id — INT, PK, AUTO_INCREMENT
- student_id — INT, NOT NULL
- section_id — INT, NOT NULL

- status — VARCHAR(20), DEFAULT 'enrolled'

Unique Constraint

- (student_id, section_id)

Indexes

- idx_student_id
- idx_section_id
- idx_status

6. grades

Columns

- grade_id — INT, PK, AUTO_INCREMENT
- enrollment_id — INT, NOT NULL
- component — VARCHAR(50)
- score — DECIMAL(7,2)
- final_grade — VARCHAR(10)

Indexes

- idx_enrollment_id
- idx_component

7. settings

Columns

- k — VARCHAR(100), PK
- v — VARCHAR(255), NOT NULL

Quick Summary

- auth_db → 2 tables
- erp_db → 7 tables
- Charset: utf8mb4
- Collation: utf8mb4_unicode_ci
- Foreign keys handled in application logic, not enforced in DB

Role & Maintenance Enforcement – Brief Summary

- Student only sees/edits own data
 - Who: StudentService, student APIs, student UI panels.
 - How:
 - AccessController.requireStudent() ensures current session user has role student.
 - Methods compare SessionManager.getCurrentUser().getUserId() to the target studentUserId and throw AccessDeniedException if they differ (e.g., “You can only register/drop for yourself.”).
 - Student APIs only return data filtered by the current user’s ID.
- Instructor only edits their sections
 - Who: InstructorService, instructor APIs, instructor UI (gradebook, section tools).

- How:
 - `AccessController.requireInstructor()` ensures role is instructor.
 - `AccessController.requireInstructorOwnsSection(sectionId, sectionDao)` checks the current instructor is assigned to that section; otherwise throws `AccessDeniedException` (“Not your section.”).
 - All write operations for rosters/grades/components go through these checks.
- Admin full control
 - Who: `AdminService`, admin APIs, `AdminDashboard`.
 - How:
 - `AccessController.requireAdmin()` at the start of admin service/API methods.
 - Admin can manage users, courses, sections, and toggle maintenance.
 - Some safety checks (e.g., cannot delete the currently logged-in admin, cannot delete instructors still assigned to sections).
- Maintenance mode → read-only for student/instructor
 - Who: `MaintenanceService`, `AccessController`, UI panels (e.g., gradebook, registrations), `MaintenanceBanner`.
 - How:
 - Global flag stored in settings; `MaintenanceService.isMaintenanceOn()` reads/caches it.
 - Write operations in services call `AccessController.requireNotInMaintenance(maintenanceService)` and fail with `AccessDeniedException` when maintenance is on.

- UI adds a red MaintenanceBanner, disables write buttons, and performs fail-fast checks (dialog “Cannot change anything – Maintenance mode is enabled”).
- Admin-only endpoints can toggle maintenance even while it is on; read operations remain allowed.

Weightage Rule

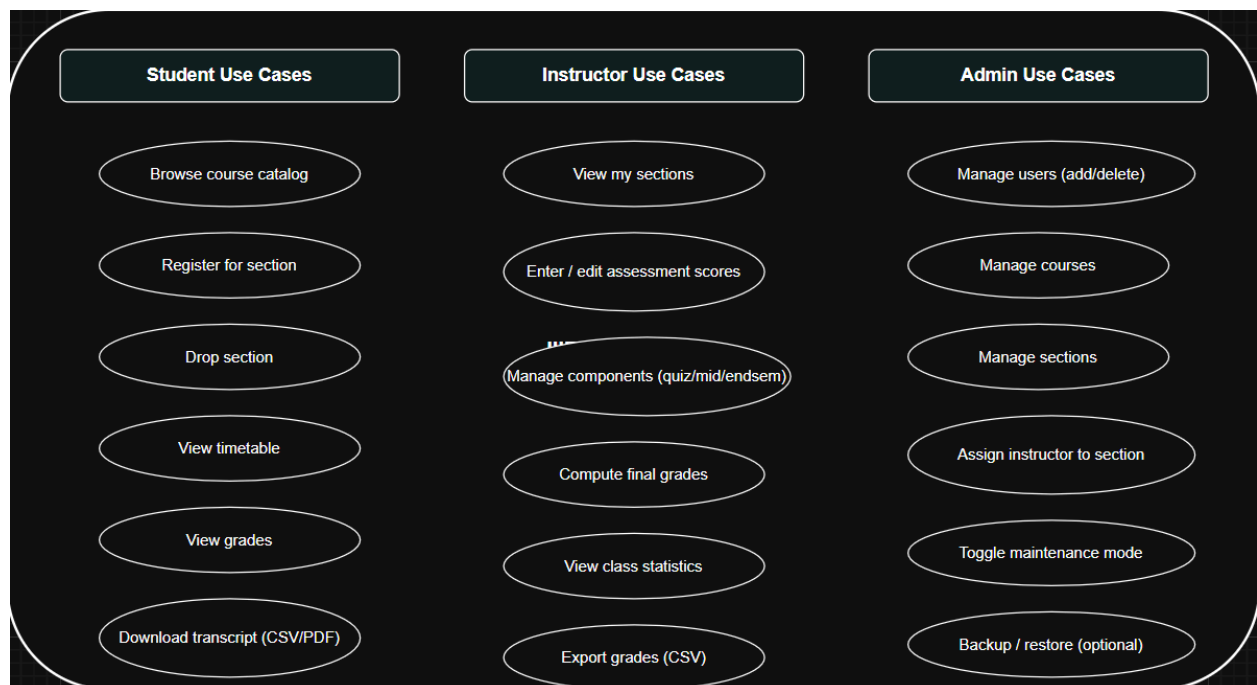
- You enter weightage of each component (quiz, midsem, endsem) and the sum must add up to 100.
- Before that, you enter each score out of hundred and save it.

Extras

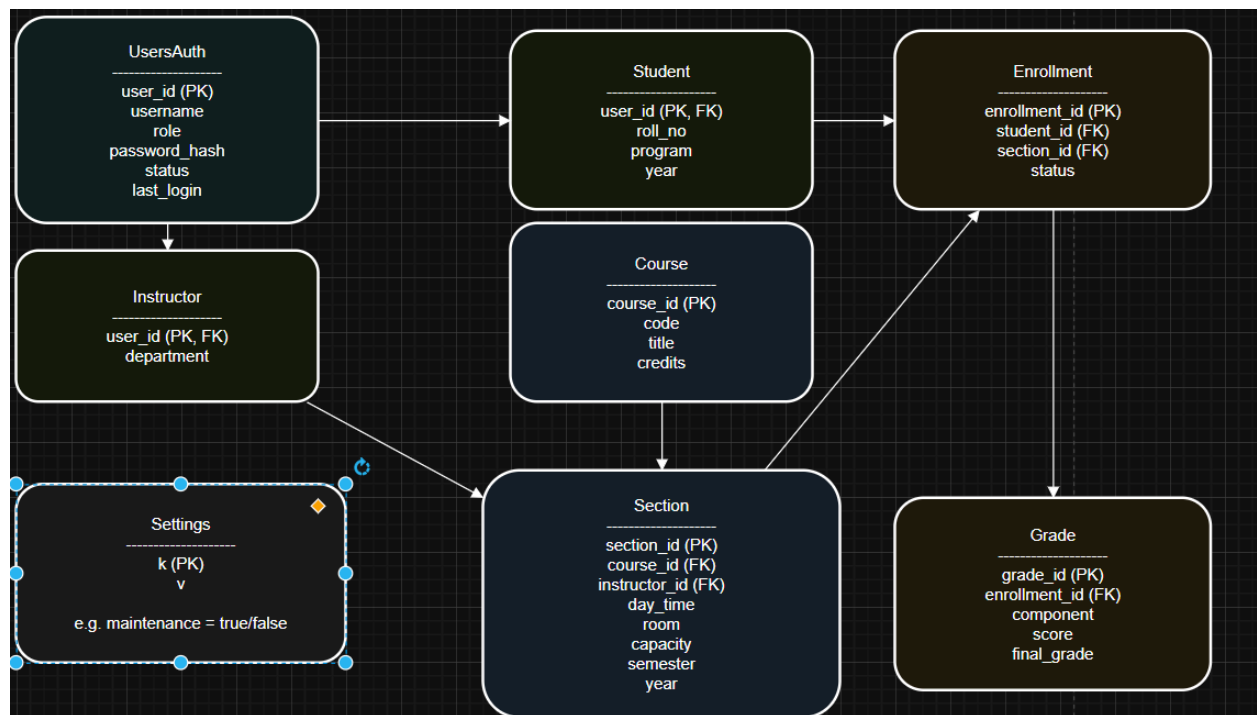
- Export csv transcript for student grades.

DIAGRAMS

Use-case list diagram



Things Sketch



Flow Sketches (enroll, grade entry, maintenance mode)

